04 / Grounding

What connects the representation or interpretation to physical evidence?

EX05 / LEAP Hand | Source analysis | C1 | P: paper; W: project; R: repository

Prediction / representation	Matched reference	Difference / ambiguity	Conclusion / scope limit

1. Comparison boundary

Name the prediction/measurement and reference. Match conditions; identify simulated versus physical evidence.

Compare Pick task success on LEAP and Allegro in the reported VideoDex workflow; keep Table IV teleoperation and Table VII simulated rotation separate. [P Table VI]

2. Quantities and validity

Declare quantity, units, frame, tolerance and measurement limits. Distinguish a chosen tolerance from observed uncertainty.

Success is a task outcome, not commanded angle, motor current or direct contact force. Test-object identity, denominator and judging detail remain NE here.

3. Results and uncertainty

Give paired outcomes, trial counts and uncertainty or limits. If untested, record the protocol and missing evidence explicitly.

Pick test fractions: 0.84 versus 0.75 (+9 percentage points). Training fractions are 0.92/0.81. Do not infer trial counts or uncertainty from decimal precision. [P Table VI]

4. Interpretation and challenge

Bound the conclusion and consider another cause of the difference. Choose a diagnostic test or scope restriction.

Morphology, retargeting and controller effects can all contribute. This full-system contrast does not isolate hardware alone or establish physical cube-rotation reliability. [P VI-C]

Review: Could sensing error, reset differences or selection explain the discrepancy instead of model error?

Example response: The nine-point test difference is a full-system contrast; no denominator is inferred.

Worksheet 4 / Grounding – evidence and team review

EX05 / LEAP Hand | SOURCE ANALYSIS EXAMPLE | Original review-page layout

Example ID: EX05 Project: LEAP Hand Claim ID: C1 Revision: draft 2

Mode: Source analysis Pages: canvas + this review; Decision also includes both A3 sides

Evidence scope: selected paper text/tables checked 20 Oct 2026; no raw dataset download or independent experiment.

Evidence ID	Artifact/source + location (clickable)	What it supports; limitation / access gap
P	arxiv.org/html/2309.06440v1	Paper: method/results; reported evidence, not our measurement.
W	v1.leaphand.com	Project: system/videos; selected examples do not establish reliability.
R	github.com/leap-hand/LEAP_Hand_API	Repository: linked route; code, raw data and current commit not audited.

Next action / owner / verification:

Analyst: verify the 0.84/0.75 Pick contrast; obtain trial count before reporting uncertainty or reliability.

Teaching example: no actual student team or signatures. Source analysis is not a reproduced result; blank signature fields are intentional.

1. Name / student ID: _____

Signature: _____ Date: _____

2. Name / student ID: _____

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8. Name / student ID: _____

Signature: _____ Date: _____

P/W/R identify linked public sources above. NE: not examined in this brief example, not evidence of failure. Support/context labels are provisional, not grades.